Boosting Regressor Algorithms

* Ensemble method
* Combines weak learners
* Improves prediction accuracy

What is Boosting?

* Sequential learning
* Focus on errors
* Each model improves previous

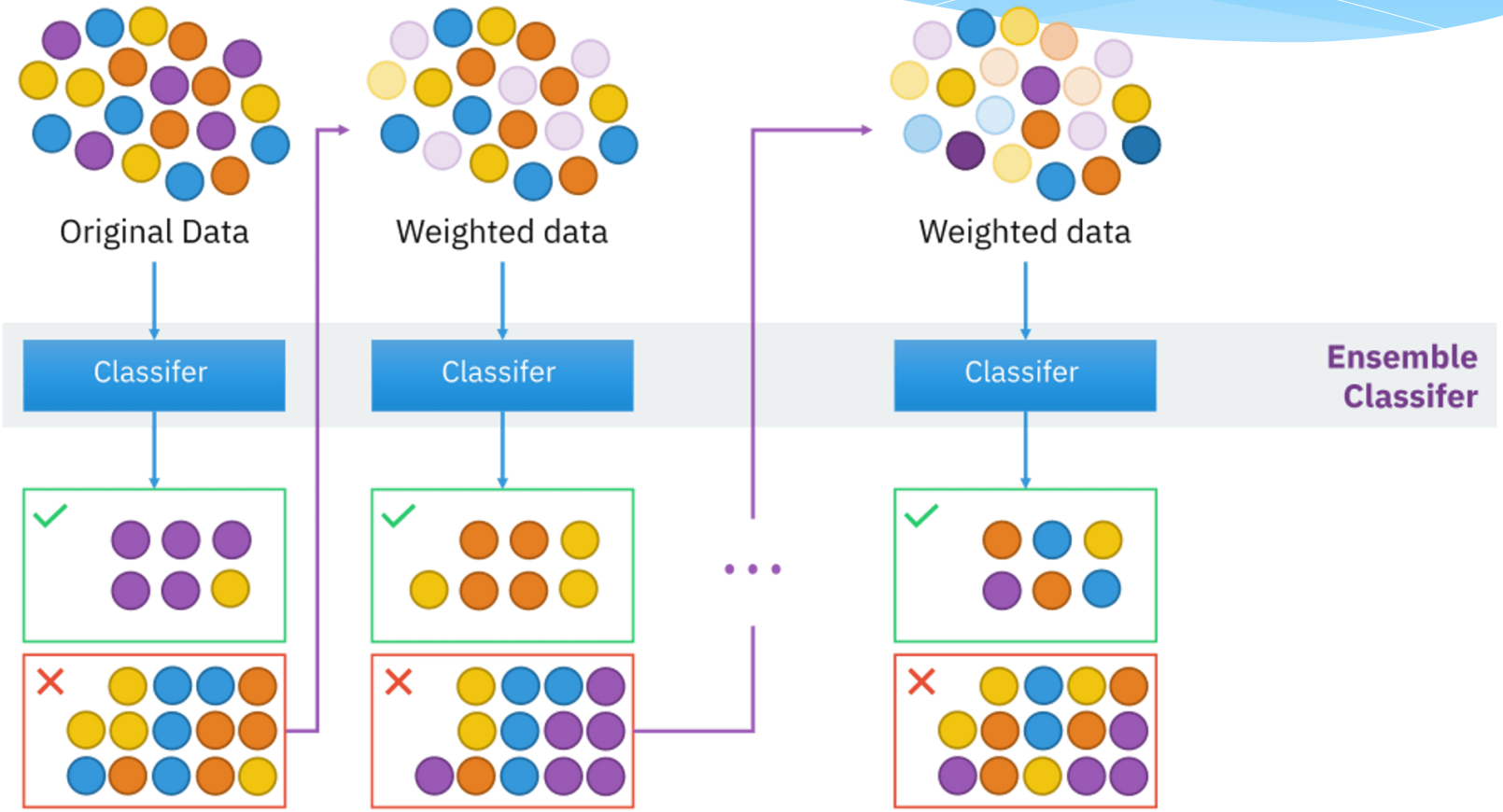
Popular Algorithms

* AdaBoost
* Gradient Boosting
* XGBoost
* LightGBM

AdaBoost Algorithms

* AdaBoost (**Ada**ptive **Boost**ing) regression is an ensemble learning method that builds a strong regressor by sequentially combining multiple "weak learners".
* In each iteration, it adjusts the weights of training instances based on their prediction error, forcing subsequent models to focus on the most difficult cases

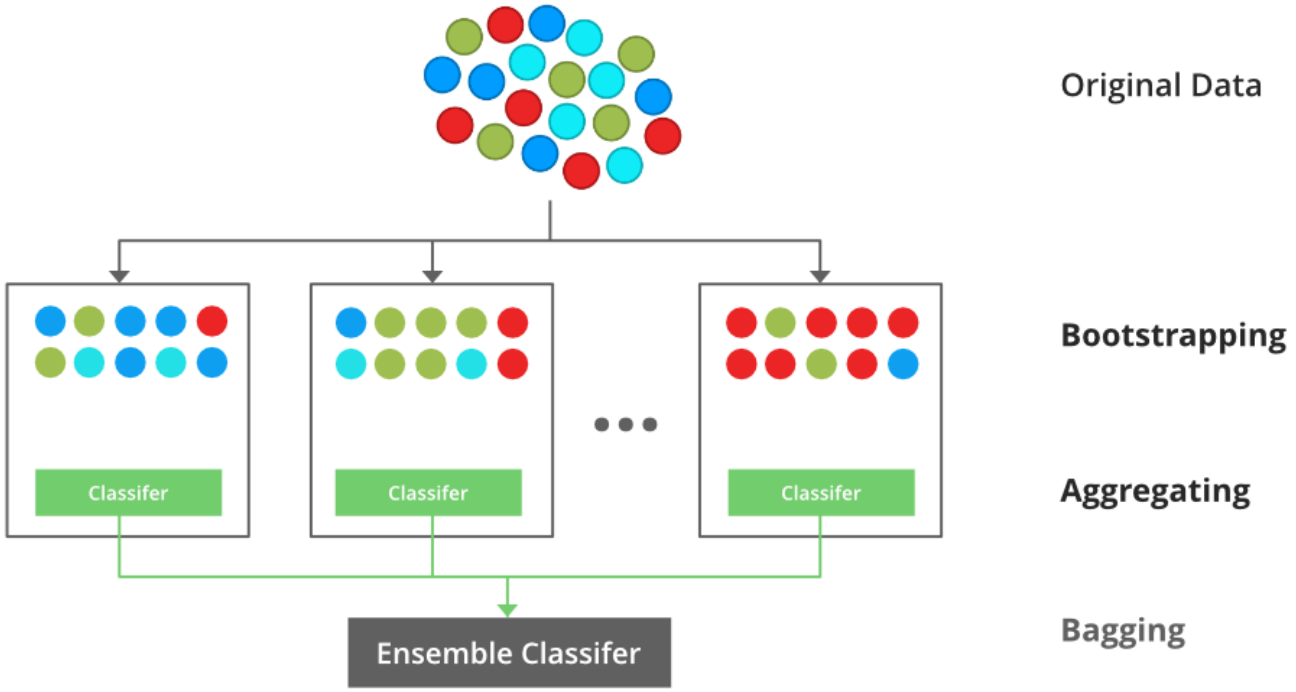
AdaBoost Algorithms



XG Boost Algorithms

* XG Boost (eXtreme Gradient Boosting) is a popular supervised-learning algorithm used for regression and classification on large datasets.
* It uses sequentially-built shallow decision trees to provide accurate results and a highly-scalable training method that avoids overfitting.

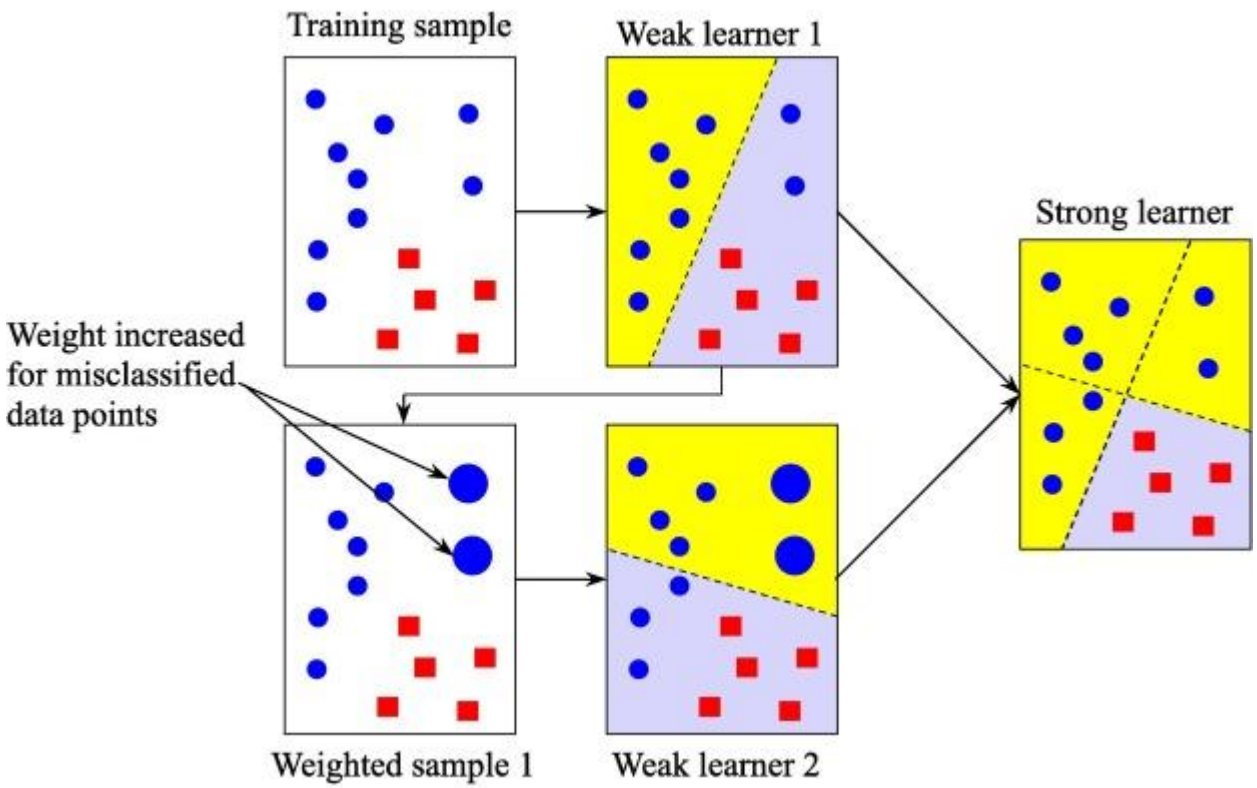
XG Boost Algorithms



LG Boost Algorithms

* LightGBM is a gradient boosting ensemble method that is used by the Train Using AutoML tool and is based on decision trees.
* It is optimized for high performance with distributed systems.
* It is widely used in Python to handle large datasets, often outperforming other boosting algorithms in terms of speed and memory usage.

LG Boost Algorithms



Advantages

* High accuracy
* Handles complex data
* Reduces bias & variance

Limitations

* Can over fit
* Slower training
* Sensitive to noise